

The Business Problem



Modern IoT environments face constant security threats, risking data loss and downtime.



Unidentified anomalies can lead to costly breaches and service interruptions.



Investing in robust detection systems protects assets and ensures business continuity.

Our Approach



Analyzed network activity logs to spot unusual patterns signaling potential threats.



Compared several detection methods to find the most reliable solution.



Focused on practical, real-time protection for diverse IoT devices.

Key Findings

Certain attack types are much more common, requiring targeted defenses.

Some network features strongly indicate suspicious activity.

Tree-based models excel at identifying threats, but rare attack types remain challenging.

Results

Best model detects threats with over 97% accuracy, minimizing false alarms.

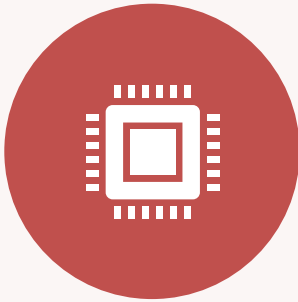


Enables proactive security, reducing risk of costly incidents.



Supports scalable protection as IoT deployments grow.

Recommendations



DEPLOY ADVANCED
ANOMALY DETECTION FOR
REAL-TIME MONITORING.



REGULARLY UPDATE
MODELS TO ADDRESS NEW
ATTACK TYPES.



BALANCE ACCURACY WITH
FAIRNESS TO AVOID
MISSING RARE THREATS.

Risks & Considerations



Rare attacks are still hard to detect—continuous improvement needed.



Overfitting to common attack types may leave gaps in protection.

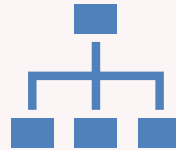


Fairness and transparency in detection must be maintained.

Next Steps & ROI Projection



Explore advanced models for even better results.



Implement hierarchical detection to focus on critical threats.



Projected ROI: Reduced incident costs, improved uptime, and enhanced customer trust.